RNNs and LSTMs

ECON 8310: Business Forecasting — Lecture 9, Part 1

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Lecture Outline

1. Where the Last Two Lectures Ran Out
2. The Vanilla RNN
3. LSTM: Long Short-Term Memory
4. In PyTorch, and Does It Work
5. Key Takeaways

Where the Last Two Lectures Ran Out

Two architectures, two different ways of failing to see time.

Two Architectures That Cannot See Time

Lecture 7's feedforward network treats each row independently: everything it knows about *when*, you encoded by hand as lag columns.

Lecture 8's 1D CNN sees local shape, but each filter's **receptive field** is about seven weeks (our two small kernels; depth or dilation would widen it), and pooling threw away *where* each pattern occurred — a measured 500 RMSE.

> Both treat the input as a **fixed-size block**, processed all at once. Neither reads a sequence *in order*, carrying something forward. That is the gap today closes.

The Recurrent Idea: Carry State Forward

A recurrent network reads the sequence one step at a time, keeping a **hidden state** $\mathbf{h}_t$: a running summary of everything so far. Each step updates the summary from the new input *and* the old summary, with the same weights every step — weight sharing again, along time instead of space.

The promise: no fixed window. In principle $\mathbf{h}_t$ carries the whole history.

In principle. Most of this lecture is about the gap between that promise and what a recurrent network delivers — and the architecture built to close it.

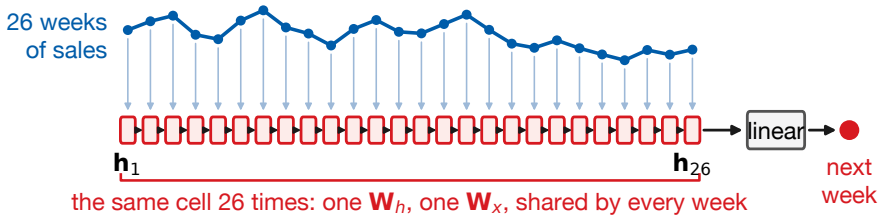
The Vanilla RNN

One equation, applied over and over — and the reason that is both the whole idea and the whole problem.

Vanilla RNN: One Equation, Repeated

The recurrence

$$\mathbf{h}_t = \tanh(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}), \qquad \hat{y}_t = \mathbf{W}_y \mathbf{h}_t + \mathbf{b}_y$$



In words: the new summary is a squashed mix of the old summary and the new week. Shared weights mean a 26-week window costs no more parameters than a 5-week one — about **4,500** here, against 22,657 for Lecture 7's network.

Training It: Backpropagation Through Time

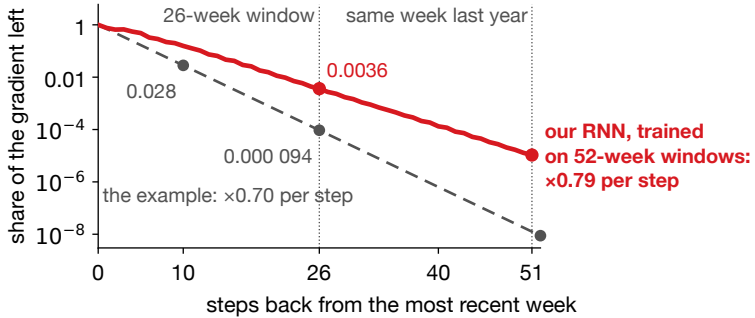
Training is ordinary backpropagation on an unusual graph. **Unroll** the recurrence and it is a 26-layer network, one layer per week, every layer sharing the same weights. Gradients flow back from step T to step 1, and $\mathbf{W}_h$'s gradient sums contributions from all 26.

> **This is where it breaks.** Going back k steps multiplies by $\mathbf{W}_h$ and by $\tanh' \leq 1$, k times. Below 1, the gradient shrinks *exponentially*; above 1, it can explode.

Exploding is fixed by `clip_grad_norm_`; vanishing is not — the next section.

Why a Vanilla RNN Forgets (Bengio et al. 1994)

Suppose each backward step multiplies the gradient by 0.7 — an unremarkable value for a $\tanh$ network — and compare our own RNN, trained on 52-week windows, measured:



The same factor every step is a straight line on a log scale. Past the first few steps, ours keeps about 0.79 a step; measured, **0.4%** of the gradient reaches 26 weeks back, **0.001%** the same week last year. **Bengio and colleagues** proved this is inherent to the architecture.

LSTM: Long Short-Term Memory

Add a memory that information passes through rather than being recomputed at every step.

LSTM: A Cell State and Three Gates

An LSTM carries **two** things forward: a **cell state** $\mathbf{c}_t$, which is the long-term memory, and a **hidden state** $\mathbf{h}_t$, which is what the rest of the network sees. Three gates control the traffic between them (Hochreiter and Schmidhuber 1997; Gers et al. 2000).

Forget gate	$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f)$ — what fraction of the old memory to keep
Input gate	$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i)$ — how much of the new candidate to write
Output gate	$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o)$ — how much of memory to expose
Candidate	$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c)$ — the new content on offer

The memory update itself is the line that matters:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} \; + \; \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \qquad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$$

Here σ is the sigmoid, so each gate outputs between 0 and 1 — a soft switch the network learns. $[\mathbf{h}_{t-1}, \mathbf{x}_t]$ stacks the two vectors; $\odot$ multiplies element by element.

What the Gates Are Doing, in Plain Terms

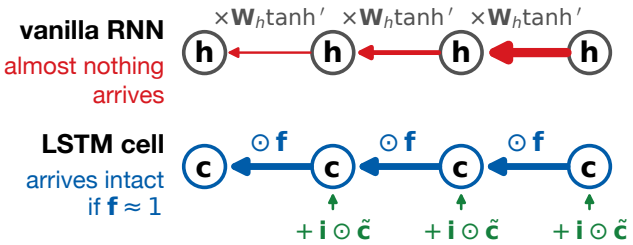
The equations are compact but the behavior is intuitive. Take a store forecasting demand week by week.

Forget	"The promotion that ran three months ago no longer explains anything — drop it." The gate closes toward 0 and that part of memory decays away.
Input	"This week is Thanksgiving and the spike is enormous — record it." The gate opens toward 1 and the event is written into the cell.
Output	"I am holding a memory of last December, but this week is April — do not act on it yet." Information stays in $\mathbf{c}_t$ without contaminating the current prediction.

The output gate is the one people skip, and it is the subtle one. It separates *what the model remembers* from *what the model uses right now*. A vanilla RNN cannot make that distinction — its single hidden state has to serve both jobs at once, so anything it stores is immediately in play.

Why the Cell State Fixes the Gradient

Follow a gradient backward through $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$. Repeated multiplication shrinks; repeated addition does not.



You have seen this fix before: Lecture 8's skip connection, $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$. The 1997 LSTM had no forget gate, so its cell path *was* that identity, eighteen years before ResNet; the forget gate (2000) made it learnable.

On our panel the median forget gate is 0.52, so 7% survives four weeks. Check 2 says why.

In PyTorch, and Does It Work

Four lines of model code, and an honest measurement against everything else this course has fitted.

LSTM for Forecasting in PyTorch

Window in, one forecast out

```
class LSTMForecaster(nn.Module):
  def __init__(self, n_features, hidden=64, layers=1):
    super().__init__()
    self.lstm = nn.LSTM(n_features, hidden, num_layers=layers,
                        batch_first=True)
    self.fc = nn.Linear(hidden, 1)
  def forward(self, x):      # x: (batch, seq_len, n_features)
    out, (h, c) = self.lstm(x)  # nn.RNN instead: out, h = self.rnn(x)
    return self.fc(out[:, -1, :]).squeeze(-1)
```

`batch_first=True` gives shape `(batch, seq_len, features)` — the *opposite* of `Conv1d`. `out[:, -1, :]` is the state after the last step. Lecture 7's rules still hold, plus `clip_grad_norm_(..., 1.0)`.

Does It Work? The Scoreboard So Far

Same weekly panel, same 26-week windows, same test block as Homework 4.

Model	Test RMSE	Parameters
LASSO, 46 engineered features	**744**	46
XGBoost, 46 engineered features	781	—
Vanilla RNN, 4 raw channels	**842**	4,545
Random forest	899	—
LSTM, 4 raw channels	987	17,985
1D CNN, best variant	1,032	8,161
Seasonal naive	2,152	0

The vanilla RNN beats convolution decisively (842 against 1,032; the LSTM only edges it) — on *four raw channels* with no lag engineering at all, against 46 hand-built features for LASSO. That is the case for reading a sequence in order, in one line.

And the LSTM lost to the vanilla RNN — at a fixed 30-epoch budget. That is not a typo, and the next four slides are about what it does and does not mean.

Why the Gated Model Lost at 30 Epochs

All three windows share one panel and one split: one finding measured three ways, not three tests.

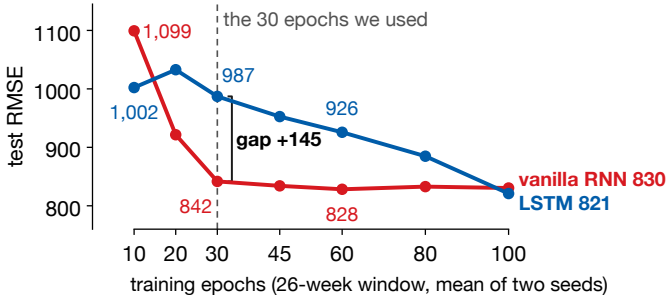
Window	Vanilla RNN	LSTM	
13 weeks	**878**	990	RNN wins
26 weeks	**842**	987	RNN wins
52 weeks	**805**	998	RNN wins

An LSTM solves a problem this data does not have: gating preserves information across *long* chains, and weekly demand is dominated by the recent past. The gates cost **17,985 parameters against 4,545**. Two checks first: **was the comparison fair**, and **what in the data explains it**?

> A margin measured at one training budget is a claim about that budget — and 30 epochs was just the default.

Check 1: Was It a Fair Fight?

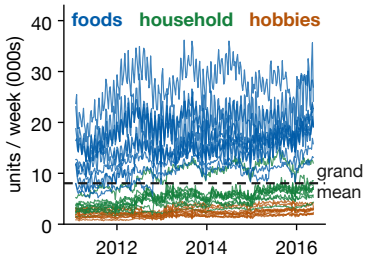
Bigger models converge slower, and the LSTM has $4\times$ the parameters. Vary the **budget**:

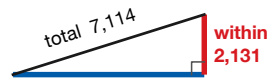


At 30 epochs the RNN wins by a wide margin. By 100 the two **tie**, and the LSTM is still improving: we ran out of patience, not the model. The RNN's real edge is speed — **three times faster on a quarter of the parameters** — not a verdict on gating.

Check 2: What in the Data Explains It

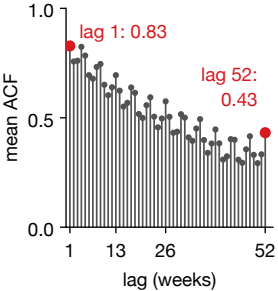
Gating pays when a value must survive many steps that would otherwise overwrite it. Does this panel have that?





between² / total² = 91%

sides: RMS deviation from the grand mean, units/week



1. Most variance is not sequential. 91% is *between* series: the task is reading the level, and the last few weeks give you that.

2. History helps, but it arrives smoothly. A longer window still improves the RNN (878 → 805), but no value has to survive many steps untouched.

Check 2, Continued: Nothing Arrives Early

3. Nothing arrives early. `snap_days` correlates $+0.14$ with units in the same week. Its -0.13 and $+0.12$ two and four weeks ahead only echo that through the monthly SNAP calendar, and `event_days` is flat everywhere. No *announcement* structure — exactly what a forget gate is for.

> Ask these of your own series before reaching for a gated model: How much of my variance is cross-sectional? Do recent values already summarize old ones? Does anything in my inputs **lead** the target?

Key Takeaways

What recurrence buys, what it costs, and what is still missing.

RNN and LSTM in Context

Architecture	Sees order?	Effective memory	Cost
FFN on lags	no	whatever you engineered	cheap
1D CNN	local only	receptive field ($\sim$7 steps)	cheap
Vanilla RNN	yes	$\sim$10–30 steps in practice	cheap
LSTM	yes	hundreds of steps	4$\times$ an RNN

An LSTM holds about **four times** an RNN's parameters at the same hidden size: three gates and the candidate each need a weight matrix. On a short window you pay that for nothing.

Still missing: an LSTM reads strictly left to right, so it cannot be parallelized, and everything about step 3 must squeeze through every state between 3 and T. Part 2 removes both.

Lecture 9, Part 1: Key Takeaways

1. A **recurrent network** reads in order, carrying a hidden state.
2. **Backpropagation through time** multiplies at every step, so the gradient shrinks exponentially — ours keeps 0.001% of it a year back.
3. An **LSTM**'s cell state is updated **additively**: the ResNet fix, eighteen years earlier.
4. Three gates: **forget** the stale, **input** the new, **output** what matters now.
5. Both **beat every CNN**; both lose to a 46-coefficient LASSO.
6. The LSTM **lost at 30 epochs and tied by 100**. Check a margin survives the training budget.

References I

- Bengio, Yoshua, Patrice Simard, and Paolo Frasconi (1994). "Learning Long-Term Dependencies with Gradient Descent Is Difficult". In: *IEEE Transactions on Neural Networks* 5.2, pp. 157–166.
- Gers, Felix A., Jürgen Schmidhuber, and Fred Cummins (2000). "Learning to Forget: Continual Prediction with LSTM". In: *Neural Computation* 12.10, pp. 2451–2471.
- Hochreiter, Sepp and Jürgen Schmidhuber (1997). "Long Short-Term Memory". In: *Neural Computation* 9.8, pp. 1735–1780.